

SciOdyssey.

Autonomous ML Research Agent

Preserve the research world. Reset the researcher.

SAME EVIDENCE. FRESH SCIENTISTS.
A RESEARCH WORLD THAT LASTS.

01 / PROBLEM

Different lifetimes. Shared evidence.

Long-lived reasoning carries yesterday’s momentum into today’s search. Tree search repeatedly rebuilds context: understanding files can cost far more tokens than editing them.

CONTEXT COST The world should outlive a single trajectory.

02 / MOTIVATION

Research world

PERSISTENT
ON DISK

EVIDENCE	Knowledge + experiment ledger Metrics, failures and original reports
CURATED	Research brief + fallible priors Current understanding, open to revision
CODE	Retained implementation + State Recoverable version, evidence and provenance

🔄 Context reset: inherit experience, not prior reasoning.

03 / SYSTEM

Scientist

SCIENTIFIC JUDGMENT

- Read the research world
- Form hypotheses
- Code, train and evaluate
- Interpret and report evidence

Chooses the science within one trajectory.

META

PERSISTENCE AUTHORITY

- Audit claims against evidence
- Scope conclusions
- Curate shared memory
- Preserve optional State

Maintains what survives across trajectories.

🔄 Audit → scope → preserve → fresh Scientist

Runtime	Isolation, cancellation, runner, evaluator and artifact capture.
Parallel breadth	Independent worktrees, followed by review and explicit adoption. A framework capability, not an isolated cause of the score.

04 / EXPERIENCE

From task to evidence.

task.md

What to solve

Objectives, constraints and evaluation

PERSONAL.md

How you want it solved

Preferences, research style and priorities

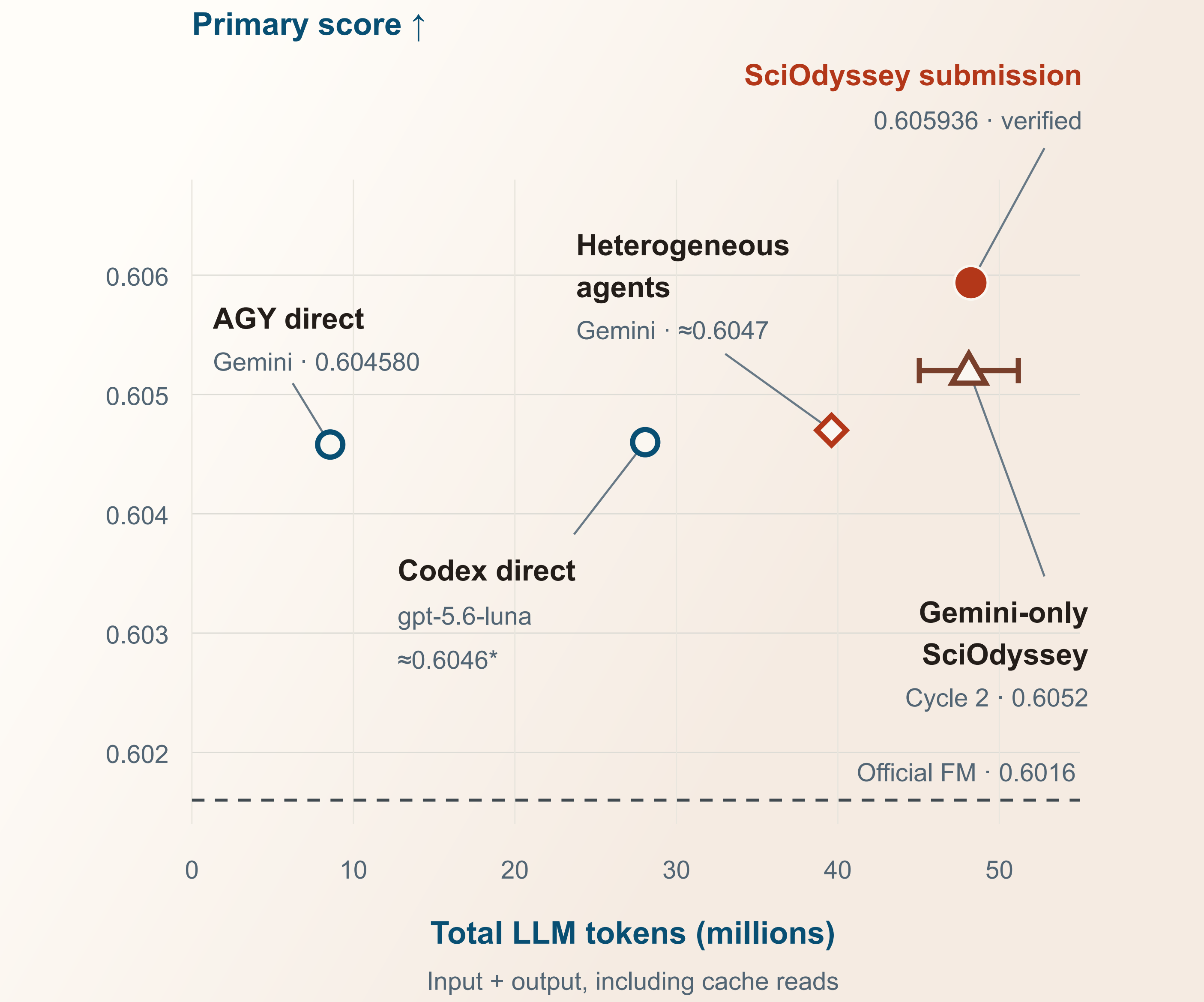
Define the task → Delegate research → Review evidence

◦ ◦ ◦ Coding agent	EXAMPLE INPUT
➤ Use the research-agent skill for task.md. Follow PERSONAL.md. Run autonomously. Let me review the results and evidence.	

05 / EVALUATION

Research outcomes, in context.

KuaiRand-Pure · public validation · one recorded run per condition



* Codex is provisional. Gemini = gemini-3.7-flash. The Cycle-2 interval is a token-accounting bound. Models and protocols differ: no causal, significance or token-efficiency claim is made.

Verified submission · public validation

Metric	Official FM	SciOdyssey	Absolute Δ
Primary	0.6016000	0.6059363	+0.0043363
GAUC	0.6674000	0.6728421	+0.0054421
nDCG@5	0.5357000	0.5390304	+0.0033304

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical features, eight-seed FM ensemble, NumPy / CPU.

4 autonomous cycles | 13 named experiments | 0 manual scientific interventions*

7 Full evaluations · 0 GPU-hours · 170,588 prediction rows passed the unchanged Starter Kit alignment checker.

* After launch, in the retained submission record.

06 / INSIGHTS

What the run taught us.

Recovery

NumPy float32 values broke serialization after evaluation. The Scientist repaired the writer and retained the evidence.

Audit

Reviews found leakage and sampling mismatches. META scopes evidence; it is not a formal verifier.

Open questions

One task does not establish generality. META, State and reset effects are not isolated. Summaries can lose context and anchor later work.

BETTER PROCESSES. A WIDER RESEARCH WORLD.

